

第十四章：数据的数字特征 20 道选择题（含习题、答案、解析）

选择题 1（平均数计算 / Mean Calculation）

某武术比赛 7 名评委给甲选手打分：90、88、93、93、92、92、96，按“去掉一个最高分、一个最低分”规则，甲选手的最后得分是（ ）

(In a martial arts competition, 7 judges scored contestant A as follows: 90, 88, 93, 93, 92, 92, 96. According to the rule of “removing one highest score and one lowest score”, what is contestant A's final score?)

A. 91 B. 92 C. 93 D. 94

答案：B (Answer: B)

解析：去掉最低分 88、最高分 96，剩余数据 90、93、93、92、92，平均数公式为 $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ ，代入得 $\bar{x} = \frac{90+93+93+92+92}{5} = 92$ 。

(Explanation: Remove the lowest score 88 and the highest score 96. The remaining data are 90, 93, 93, 92, 92. The mean formula is $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, substituting the data gives $\bar{x} = \frac{90+93+93+92+92}{5} = 92$.)

选择题 2（中位数计算 / Median Calculation）

数据 0、0、1、2、3、12 的中位数是（ ）

(What is the median of the data set: 0, 0, 1, 2, 3, 12?)

A. 1 B. 1.5 C. 2 D. 3

答案：B (Answer: B)

解析：数据个数为偶数（ $n = 6$ ），中位数公式为 $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$ ，代入得 $\text{Median} = \frac{x_3 + x_4}{2} = \frac{1+2}{2} = 1.5$ 。

(Explanation: The number of data points is even ($n = 6$). The median formula is $\text{Median} =$

$\frac{x_n + x_{n+1}}{2}$, substituting the data gives Median = $\frac{x_3 + x_4}{2} = \frac{1+2}{2} = 1.5$.)

选择题 3 (众数识别 / Mode Identification)

某班级双肩包容量意向数据：28、29、29、30、29、28、29、31、29、27、29、30，这组数据的众数是 ()

(The data on backpack capacity preferences of a class are: 28, 29, 29, 30, 29, 28, 29, 31, 29, 27, 29, 30. What is the mode of this data set?)

A. 28 B. 29 C. 30 D. 31

答案：B (Answer: B)

解析：众数是一组数据中出现次数最多的数值，29 出现 5 次，为出现次数最多的数值。

(Explanation: The mode is the most frequently occurring value in a data set. The value 29 appears 5 times, which is the most frequent.)

选择题 4 (百分位数计算 / Percentile Calculation)

数据 1、2、3、4、5、6、7、8、9、10 的 25%分位数是 ()

(What is the 25th percentile of the data set: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10?)

A. 2 B. 2.5 C. 3 D. 3.5

答案：B (Answer: B)

解析：设数据个数为 $n = 10$ ，百分位数位置公式为 $i = n \times p$ (p 为百分位)，代入得 $i = 10 \times 0.25 = 2.5$ (非整数)。按规范，非整数时取相邻两个数据的平均值，即 $\frac{x_2 + x_3}{2} = \frac{2+3}{2} = 2.5$ ，故选 B。

(Explanation: Let the number of data points be $n = 10$. The percentile position formula is $i = n \times p$ (p is the percentile). Substituting the data gives $i = 10 \times 0.25 = 2.5$ (not an integer).

According to the standard, when i is not an integer, take the average of the two adjacent data points, i.e., $\frac{x_2+x_3}{2} = \frac{2+3}{2} = 2.5$. So the answer is B.)

选择题 5 (极差计算 / Range Calculation)

数据 90、92、92、93、93 的极差是 ()

(What is the range of the data set: 90, 92, 92, 93, 93?)

A. 2 B. 3 C. 4 D. 5

答案: B (Answer: B)

解析: 极差公式为 $\text{Range} = \max(x_i) - \min(x_i)$, 代入得 $\text{Range} = 93 - 90 = 3$ 。

(Explanation: The range formula is $\text{Range} = \max(x_i) - \min(x_i)$, substituting the data gives $\text{Range} = 93 - 90 = 3$.)

选择题 6 (方差计算 / Variance Calculation)

数据 0、2、2、3、3 的方差是 ()

(What is the variance of the data set: 0, 2, 2, 3, 3?)

A. 0.8 B. 1.2 C. 1.6 D. 2.0

答案: B (Answer: B)

解析: 首先计算平均数 $\bar{x} = \frac{0+2+2+3+3}{5} = 2$, 方差公式为 $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$, 代入得:

$$s^2 = \frac{(0-2)^2 + (2-2)^2 + (2-2)^2 + (3-2)^2 + (3-2)^2}{5} = \frac{4+0+0+1+1}{5} = 1.2$$

(Explanation: First calculate the mean $\bar{x} = \frac{0+2+2+3+3}{5} = 2$. The variance formula is $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$, substituting the data gives:

$$s^2 = \frac{(0-2)^2 + (2-2)^2 + (2-2)^2 + (3-2)^2 + (3-2)^2}{5} = \frac{4+0+0+1+1}{5} = 1.2$$

选择题 7 (线性变换后的平均数 / Mean after Linear Transformation)

已知数据 x_1, x_2, x_3 的平均数为 5, 若 $y_i = 2x_i + 3$, 则 y_1, y_2, y_3 的平均数是 () (Given that the mean of data x_1, x_2, x_3 is 5. If $y_i = 2x_i + 3$, what is the mean of y_1, y_2, y_3 ?)

A. 10 B. 13 C. 15 D. 18

答案: B (Answer: B)

解析: 根据平均数线性变换性质 $\bar{y} = a\bar{x} + b$ (其中 $y_i = ax_i + b$), 代入 $a = 2, b = 3, \bar{x} = 5$, 得 $\bar{y} = 2 \times 5 + 3 = 13$ 。(Explanation: According to the property of linear transformation of the mean $\bar{y} = a\bar{x} + b$ (where $y_i = ax_i + b$), substituting $a = 2, b = 3$, and $\bar{x} = 5$ gives $\bar{y} = 2 \times 5 + 3 = 13$.)

选择题 8 (线性变换后的方差 / Variance after Linear Transformation)

已知数据 x_1, x_2, x_3 的方差为 4, 若 $y_i = 3x_i - 2$, 则 y_1, y_2, y_3 的方差是 () (Given that the variance of data x_1, x_2, x_3 is 4. If $y_i = 3x_i - 2$, what is the variance of y_1, y_2, y_3 ?)

A. 4 B. 12 C. 36 D. 40

答案: C (Answer: C)

解析: 根据方差线性变换性质 $s_y^2 = a^2 s_x^2$ (其中 $y_i = ax_i + b$, a 为 x_i 的系数), 代入 $a = 3, s_x^2 = 4$, 得 $s_y^2 = 3^2 \times 4 = 36$ 。(Explanation: According to the property of linear transformation of variances $s_y^2 = a^2 s_x^2$ (where $y_i = ax_i + b$ and a is the coefficient of x_i), substituting $a = 3$ and $s_x^2 = 4$ gives $s_y^2 = 3^2 \times 4 = 36$.)

选择题 9 (求和符号计算 / Summation Symbol Calculation)

已知 $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$, 则 $\sum_{i=1}^5 x_i$ 的值是 () (Given $x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5$, what is the value of $\sum_{i=1}^5 x_i$?)

A. 6 B. 9 C. 12 D. 15

答案：D (Answer: D)

解析：求和符号定义为 $\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n$ ，代入得 $\sum_{i=1}^5 x_i = 1 + 2 + 3 + 4 + 5 = 15$ 。

(Explanation: The summation symbol is defined as $\sum_{i=1}^n x_i = x_1 + x_2 + \cdots + x_n$. Substituting the data gives $\sum_{i=1}^5 x_i = 1 + 2 + 3 + 4 + 5 = 15$.)

选择题 10 (数据特征的优缺点 / Advantages and Disadvantages of Data Characteristics)

下列关于平均数的说法，错误的是 () (Which of the following statements about the mean is incorrect?)

A. 利用所有数据信息 (Uses all data information) B. 受极端值影响显著 (Significantly affected by outliers) C. 适合偏态分布数据 (Suitable for skewed distribution data) D. 计算简单 (Simple to calculate)

答案：C (Answer: C)

解析：平均数对偏态分布数据的中心位置刻画不准确，适合对称分布数据（如正态分布 $N(\mu, \sigma^2)$ ）。(Explanation: The mean cannot accurately describe the central tendency of skewed distribution data; it is suitable for symmetric distribution data (e.g., normal distribution $N(\mu, \sigma^2)$)).

选择题 11 (中位数的优点 / Advantages of Median)

中位数的核心优点是 () (What is the core advantage of the median?)

A. 利用所有数据信息 (Uses all data information) B. 不受极端值影响 (Not affected by outliers) C. 适合分类数据 (Suitable for categorical data) D. 计算复杂 (Complex to calculate)

答案：B (Answer: B)

解析：中位数仅依赖中间位置数据，其计算公式为： - 当 n 为奇数时， $\text{Median} = \frac{x_{\frac{n+1}{2}}}{2}$ ； - 当 n 为偶数时， $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$ ， 不受极端值影响，稳定性强。（Explanation: The median only depends on the data at the middle position. Its calculation formula is: - When n is odd, $\text{Median} = \frac{x_{\frac{n+1}{2}}}{2}$; - When n is even, $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$. It is not affected by outliers and has strong stability.)

选择题 12（四分位数间距 / Interquartile Range (IQR)）

数据 1、2、3、4、5、6 的第一四分位数 $Q_1 = 2$ ，第三四分位数 $Q_3 = 5$ ，则四分位数间距 (IQR) 是 () (For the data set 1, 2, 3, 4, 5, 6, the first quartile $Q_1 = 2$ and the third quartile $Q_3 = 5$. What is the interquartile range (IQR)?)

A. 2 B. 3 C. 4 D. 5

答案：B (Answer: B)

解析：四分位数间距公式为 $\text{IQR} = Q_3 - Q_1$ ，代入得 $\text{IQR} = 5 - 2 = 3$ 。（Explanation: The interquartile range formula is $\text{IQR} = Q_3 - Q_1$, substituting the data gives $\text{IQR} = 5 - 2 = 3$.)

选择题 13（标准差的意义 / Meaning of Standard Deviation）

标准差为 0 的一组数据，说明 () (If the standard deviation of a data set is 0, it indicates that?)

A. 数据全为 0 (All data are 0) B. 数据无波动 (所有数据相等) (No data fluctuation (all data are equal)) C. 平均数为 0 (The mean is 0) D. 极差为 0 (The range is 0)

答案：B (Answer: B)

解析：标准差公式为 $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$ ，当 $s = 0$ 时， $\sum_{i=1}^n (x_i - \bar{x})^2 = 0$ ，即所有 $x_i = \bar{x}$ ，数据无波动。（Explanation: The standard deviation formula is $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$. When $s = 0$, $\sum_{i=1}^n (x_i - \bar{x})^2 = 0$, which means all $x_i = \bar{x}$ and there is no data fluctuation.)

选择题 14 (数据分布与特征的关系 / Relationship Between Data Distribution and Characteristics)

在右偏分布数据中，下列关系正确的是 ()

(In a right-skewed distribution, which of the following relationships is correct?)

- A. 平均数>中位数>众数 (Mean > Median > Mode)
- B. 中位数>平均数>众数 (Median > Mean > Mode)
- C. 众数>中位数>平均数 (Mode > Median > Mean)
- D. 平均数=中位数=众数 (Mean = Median = Mode)

答案: A (Answer: A)

解析: 右偏分布中，极端大值拉高平均数，使得平均数向右侧偏移，而中位数仅受中间位置数据影响，众数是出现次数最多的数值（不受极端值影响），因此三者关系为 $\text{Mean} > \text{Median} > \text{Mode}$ 。

(Explanation: In a right-skewed distribution, extreme large values pull up the mean, causing the mean to shift to the right. The median is only affected by the data at the middle position, and the mode is the most frequently occurring value (not affected by outliers). Therefore, the relationship among the three is $\text{Mean} > \text{Median} > \text{Mode}$.)

选择题 15 (Excel 函数应用 / Application of Excel Functions)

在 Excel 中，计算数据 A2:A8 的中位数，应使用的函数是 ()

(In Excel, which function should be used to calculate the median of data in the range A2:A8?)

- A. AVERAGE(A2:A8) (Calculates the mean)
- B. MEDIAN(A2:A8) (Calculates the median)
- C. MODE.SNGL(A2:A8) (Calculates the mode)

D. MAX(A2:A8) (Calculates the maximum value)

答案：B (Answer: B)

解析：MEDIAN 函数用于计算中位数，其核心逻辑与数学定义一致：当数据个数 n 为奇数时，返回第 $\frac{n+1}{2}$ 个数据；当 n 为偶数时，返回第 $\frac{n}{2}$ 个与第 $\frac{n}{2} + 1$ 个数据的平均值。AVERAGE 是平均数函数，MODE.SNGL 是众数函数，MAX 是最大值函数。

(Explanation: The MEDIAN function is used to calculate the median, and its core logic is consistent with the mathematical definition: when the number of data points n is odd, it returns the $\frac{n+1}{2}$ -th data; when n is even, it returns the average of the $\frac{n}{2}$ -th and $\frac{n}{2} + 1$ -th data. AVERAGE is the mean function, MODE.SNGL is the mode function, and MAX is the maximum value function.)

选择题 16 (众数的特点 / Characteristics of Mode)

下列关于众数的说法，正确的是 ()

(Which of the following statements about the mode is correct?)

- A. 一定存在且唯一 (Must exist and be unique)
- B. 不受极端值影响 (Not affected by outliers)
- C. 利用所有数据信息 (Uses all data information)
- D. 适合连续数据 (Suitable for continuous data)

答案：B (Answer: B)

解析：众数是一组数据中出现次数最多的数值，其确定仅依赖数据的出现频率，与极端值无关 (B 正确)；众数可能不存在 (如所有数据出现次数相同) 或不唯一 (如多个数据出现次数相同且最多，即多众数) (A 错误)；众数仅反映出现频率最高的数据特征，未利用所有数据信息 (C 错误)；连续数据的取值通常唯一，难以出现重复值，因此众数更适合分类数据或离散数据 (D 错误)。

错误)。

(Explanation: The mode is the most frequently occurring value in a data set. Its determination only depends on the frequency of data occurrence and is independent of outliers (B is correct); the mode may not exist (e.g., all data appear the same number of times) or be non-unique (e.g., multiple data appear the same maximum number of times, i.e., multimodal) (A is incorrect); the mode only reflects the characteristic of the most frequent data and does not use all data information (C is incorrect); the values of continuous data are usually unique, making it difficult to have repeated values, so the mode is more suitable for categorical data or discrete data (D is incorrect).)

选择题 17 (百分位数定义 / Definition of Percentile)

一组数据的 90%分位数, 满足的条件是 ()

(The 90th percentile of a data set satisfies the condition that?)

- A. 90%的数据大于该值 (90% of the data are greater than this value)
- B. 至少 90%的数据不大于该值 (At least 90% of the data are not greater than this value)
- C. 50%的数据等于该值 (50% of the data are equal to this value)
- D. 10%的数据不小于该值 (10% of the data are not less than this value)

答案: B (Answer: B)

解析: 百分位数的严格定义为: 对于一组有序数据, 第 p 百分位数 (记为 P_p) 是满足以下两个条件的数值: ①至少 p 的数据小于或等于 P_p (即不大于 P_p); ②至少 $(100 - p)$ 的数据大于或等于 P_p (即不小于 P_p)。因此 90%分位数需满足至少 90%的数据不大于该值, 至少 10%的数据不小于该值 (B 正确, D 错误); A 选项混淆了“大于”与“不小于”的概念, C 选项与百分位数定义无关。

(Explanation: The strict definition of a percentile is: for a set of ordered data, the p -th percentile (denoted as P_p) is a value that satisfies two conditions: ① At least p of the data are less than or equal to P_p (i.e., not greater than P_p); ② At least $(100 - p)$ of the data are greater than or equal to P_p (i.e., not less than P_p). Therefore, the 90th percentile must satisfy that at least 90% of the data are not greater than this value, and at least 10% of the data are not less than this value (B is correct, D is incorrect); Option A confuses the concepts of “greater than” and “not less than”, and Option C is irrelevant to the definition of percentile.)

选择题 18 (平均数与中位数的比较 / Comparison Between Mean and Median)

某组数据包含极端大值，下列说法正确的是 ()

(If a data set contains extreme large values, which of the following statements is correct?)

- A. 平均数比中位数更适合刻画中心位置 (The mean is more suitable than the median for describing the central tendency)
- B. 中位数比平均数更适合刻画中心位置 (The median is more suitable than the mean for describing the central tendency)
- C. 两者无差异 (There is no difference between the two)
- D. 众数比两者都不适合 (The mode is less suitable than both)

答案: B (Answer: B)

解析: 平均数的计算公式为 $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$ ，其数值会被极端大值显著拉高，导致无法准确反映数据的中心位置；而中位数仅依赖中间位置数据，计算公式为：

- 当 n 为奇数时， $\text{Median} = x_{\frac{n+1}{2}}$ ；

- 当 n 为偶数时, $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$,

不受极端值影响, 能更稳定地刻画数据的中心趋势 (B 正确, A、C 错误); 众数在某些情况下 (如数据集中且无极端值干扰) 也可用于刻画中心位置, 不能直接判定其比两者都不适合 (D 错误)。

(Explanation: The mean is calculated by $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, and its value will be significantly pulled up by extreme large values, making it unable to accurately reflect the central position of the data; the median only depends on the data at the middle position, and its calculation formula is:

- When n is odd, $\text{Median} = \frac{x_{\frac{n+1}{2}}}{2}$;
- When n is even, $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$.

It is not affected by outliers and can more stably describe the central tendency of the data (B is correct, A and C are incorrect); the mode can also be used to describe the central position in some cases (e.g., the data are concentrated and not disturbed by outliers), so it cannot be directly determined that it is less suitable than both (D is incorrect).)

选择题 19 (方差公式 / Variance Formula)

数据 x_1 、 x_2 、 ...、 x_n 的方差公式是 ()

(What is the variance formula for data x_1 , x_2 , ..., x_n ?)

- A. $\frac{1}{n} \sum_{i=1}^n x_i$ (Mean formula)
- B. $\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ (Variance formula)
- C. $\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$ (Standard deviation formula)
- D. $\max(x_i) - \min(x_i)$ (Range formula)

答案: B (Answer: B)

解析: 方差是衡量数据离散程度的统计量, 其核心是计算各数据与平均数的偏差平方的平均值,

公式为 $s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ (B 正确); A 选项是平均数公式, C 选项是标准差公式 (方差的算术平方根), D 选项是极差公式。

(Explanation: Variance is a statistic that measures the degree of data dispersion. Its core is to calculate the average of the squared deviations of each data from the mean, with the formula $s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ (B is correct); Option A is the mean formula, Option C is the standard deviation formula (the arithmetic square root of variance), and Option D is the range formula.)

选择题 20 (综合应用 / Comprehensive Application)

某班语文成绩: 76、86、74、82、77、68、62、82、72、82, 下列说法错误的是 ()

(The Chinese scores of a class are: 76, 86, 74, 82, 77, 68, 62, 82, 72, 82. Which of the following statements is incorrect?)

- A. 众数是 82 (The mode is 82)
- B. 中位数是 77 (The median is 77)
- C. 平均数是 77 (The mean is 77)
- D. 极差是 24 (The range is 24)

答案: B (Answer: B)

解析:

1. 众数: 82 出现 3 次, 是出现次数最多的数值 (A 正确);
2. 中位数: 将数据排序为 62、68、72、74、76、77、82、82、82、86, 数据个数 $n = 10$ (偶数), 中位数公式为 $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} = \frac{x_5 + x_6}{2} = \frac{76+77}{2} = 76.5$ (B 错误);
3. 平均数: $\bar{x} = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{62+68+72+74+76+77+82+82+82+86}{10} = 77$ (C 正确);
4. 极差: $\text{Range} = \max(x_i) - \min(x_i) = 86 - 62 = 24$ (D 正确)。

(Explanation:

1. Mode: The value 82 appears 3 times, which is the most frequent value (A is correct);
2. Median: Sort the data as 62, 68, 72, 74, 76, 77, 82, 82, 82, 86. The number of data points $n = 10$ (even). The median formula is $\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2} = \frac{x_5 + x_6}{2} = \frac{76 + 77}{2} = 76.5$ (B is incorrect);
3. Mean: $\bar{x} = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{62+68+72+74+76+77+82+82+82+86}{10} = 77$ (C is correct);
4. Range: $\text{Range} = \max(x_i) - \min(x_i) = 86 - 62 = 24$ (D is correct.)